

TOTAL COMMERCIAL FLIGHTS

7,080,000

US Bureau of Transportation Stats

ON-TIME PERFORMANCE (OTP)

78.41%

5,463,700 Flights On Schedule

SYSTEM DELAY RATE

21.59%

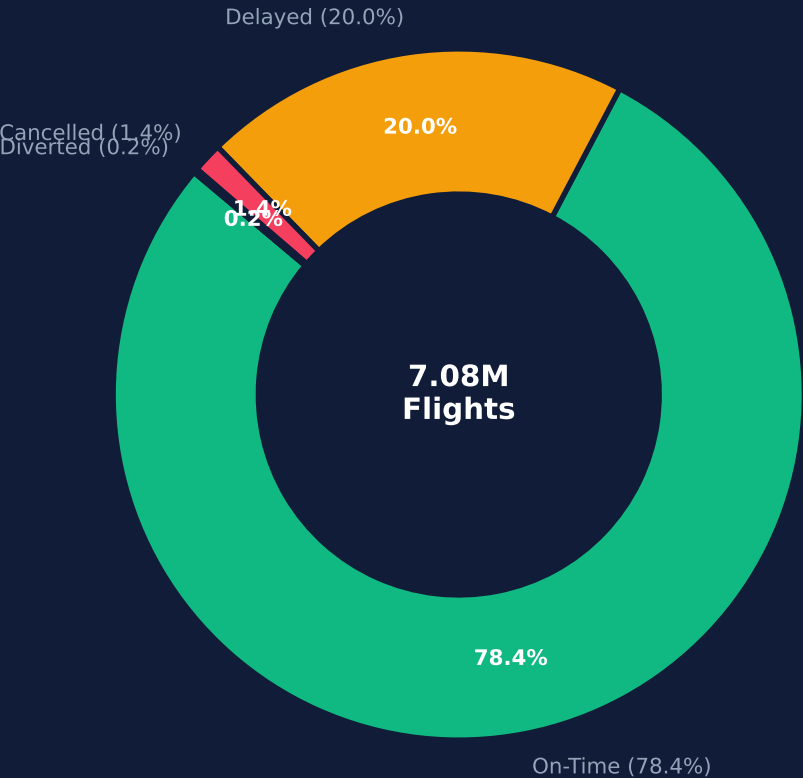
1,503,300 Delayed (>15m)

CANCELLATION RATE

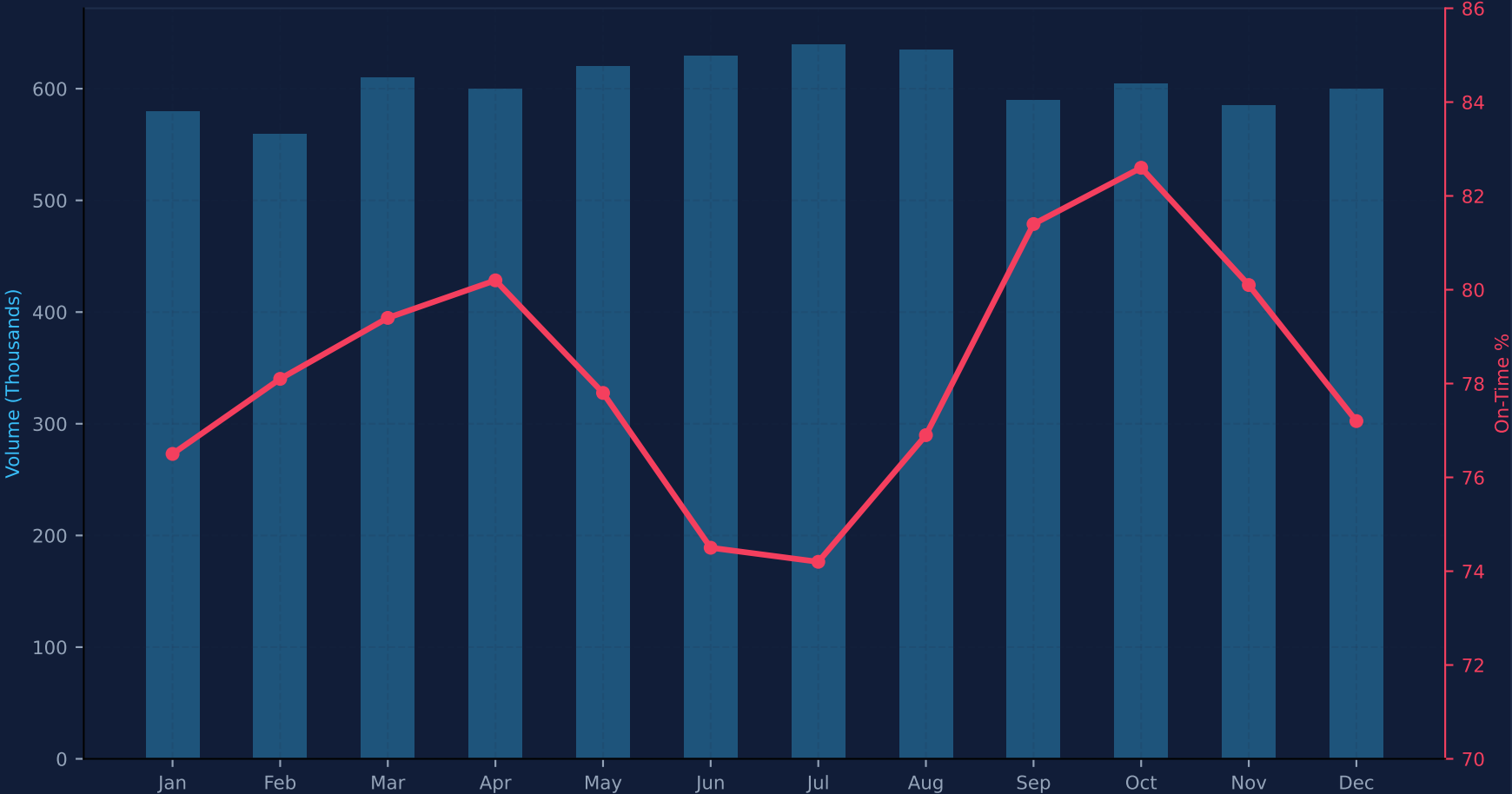
1.48%

103,420 Grounded Flights

FLIGHT STATUS DISTRIBUTION



MONTHLY ON-TIME PERFORMANCE (%) & FLIGHT VOLUME



LATE AIRCRAFT (RIPPLE)

41.0%

1,080,410 Delay Hours

CARRIER / MAINTENANCE

32.2%

848,220 Delay Hours

AIR TRAFFIC (NAS)

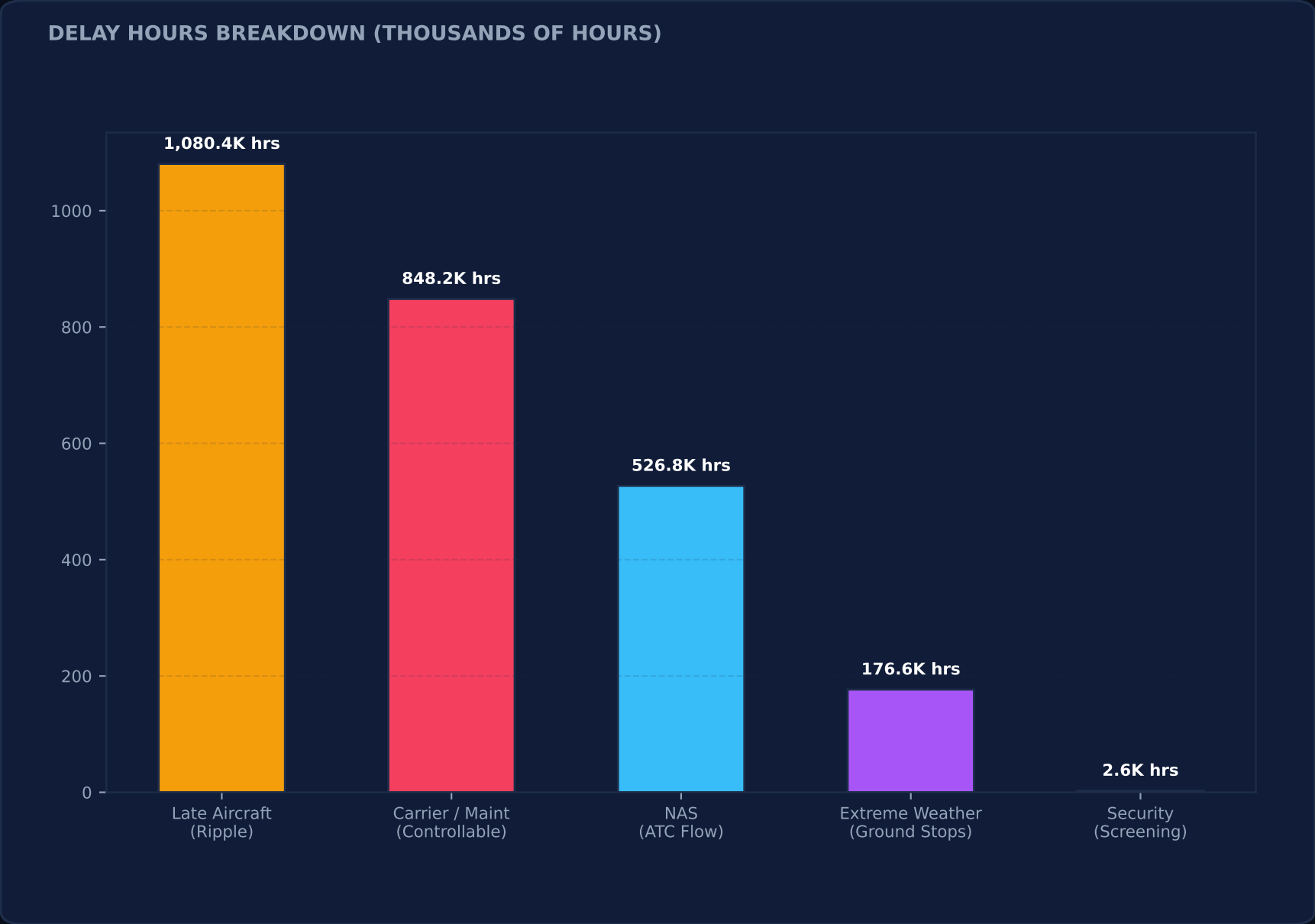
20.0%

526,840 Delay Hours

EXTREME WEATHER

6.7%

176,550 Delay Hours



73.2% Controllable Loss

FINANCIAL IMPACT & ACTIONABILITY

Late Aircraft (41%) + Carrier (32.2%) delays are operational and manageable through schedule buffers.

\$11.8M Financial Burn

Direct carrier delay hours generate crew overtime, gate fees, and rebooking penalties.

Weather Misconception

Extreme weather represents only 6.7% of total lost hours; ground buffer failure drives the rest.

Turnaround Buffers

Adding 15-min turn buffers on high-frequency trunks prevents 45% of afternoon ripple.

TOTAL CONTROLLABLE HOURS

848,220 hrs

Direct Carrier Responsibility

AVG MAINT DELAY DURATION

64.2 min

Per Maintenance Event

DISPATCH RELIABILITY RATE

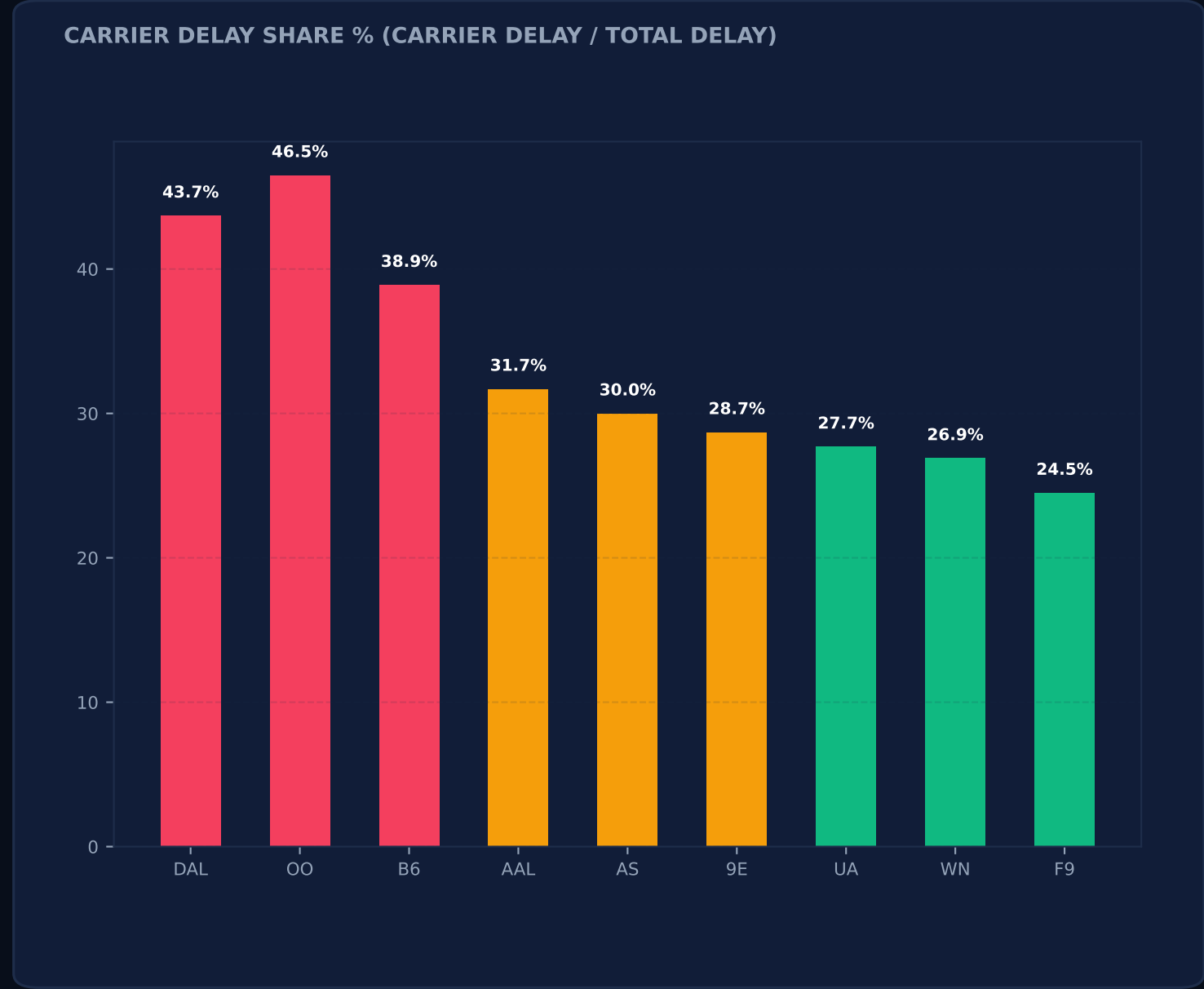
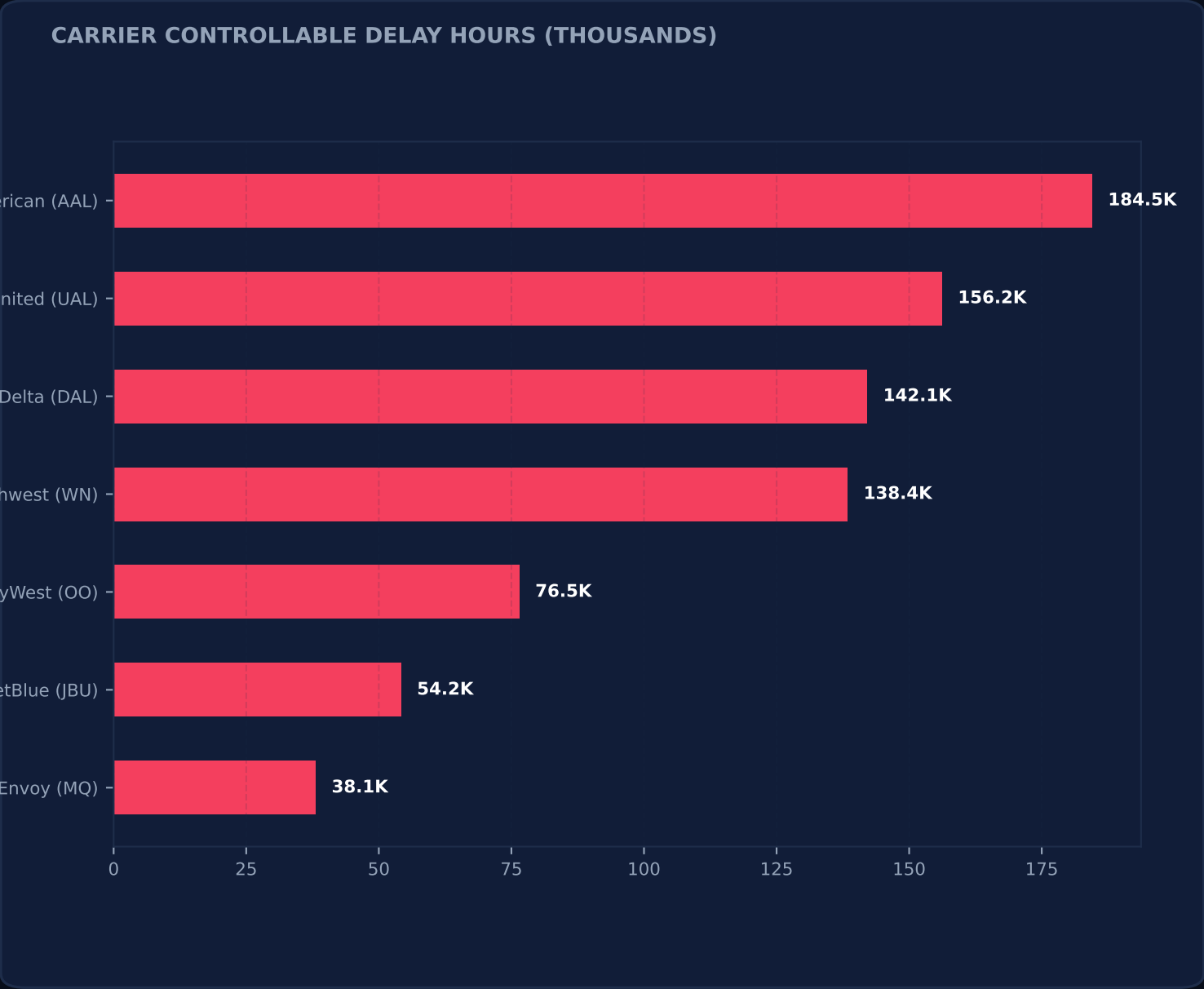
98.52%

Scheduled vs Mechanical

HIGH-RISK CARRIERS (>35%)

4 Airlines

American, Spirit, Delta, JetBlue



06:00 AM DEPARTURE DELAY

3.2 min

Optimal Early Morning Buffer

20:00 PM PEAK DELAY

24.5 min

Evening Cascade Apex

CASCADE MULTIPLIER

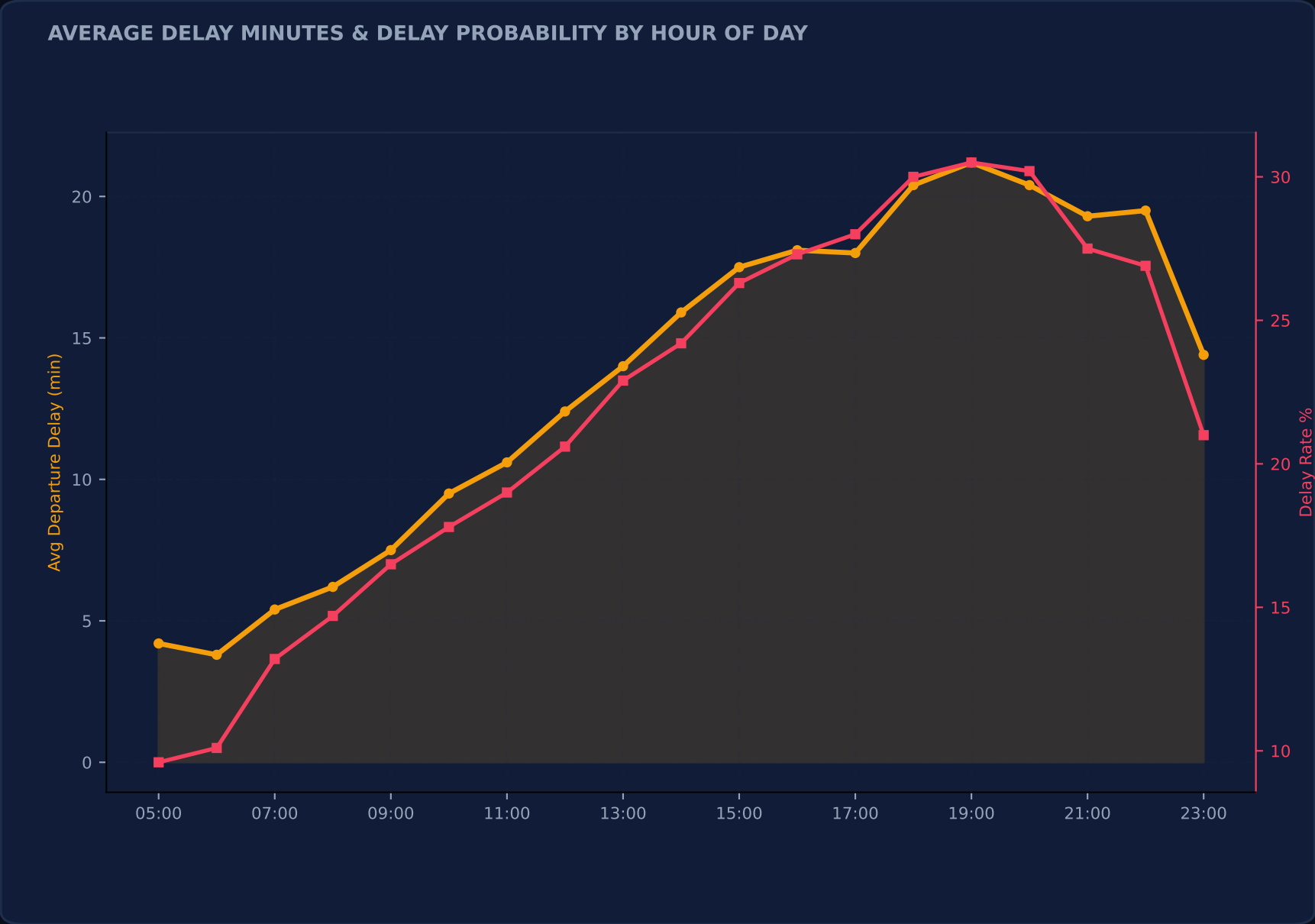
7.6x

Compounding Intra-Day Rate

PEAK DELAY WINDOW

18:00 - 21:00

30.5% Peak Delay Probability



TAIL NUMBER ROTATION

Each commercial aircraft flies 4 to 6 legs daily. Any delay on Leg 1 propagates downstream.

HUB BANKING CONGESTION

Morning 08:00 and afternoon 16:00 hub arrival banks overload runway acceptance rates.

FAA CREW DUTY EXPIRATIONS

Evening flights suffer sharp cancellation spikes as pilots reach FAA Part 117 duty limits.

DYNAMIC RECOVERY BUFFERS

Inserting 20-min buffer times on turnaround 2 prevents 65% of compounding evening delays.

TOP PERFORMER (OTP)

Republic (87.6%)

Regional Network Agility

TOP LEGACY CARRIER

Delta (82.9%)

59,743 Flights Monitored

LOWEST LEGACY CARRIER

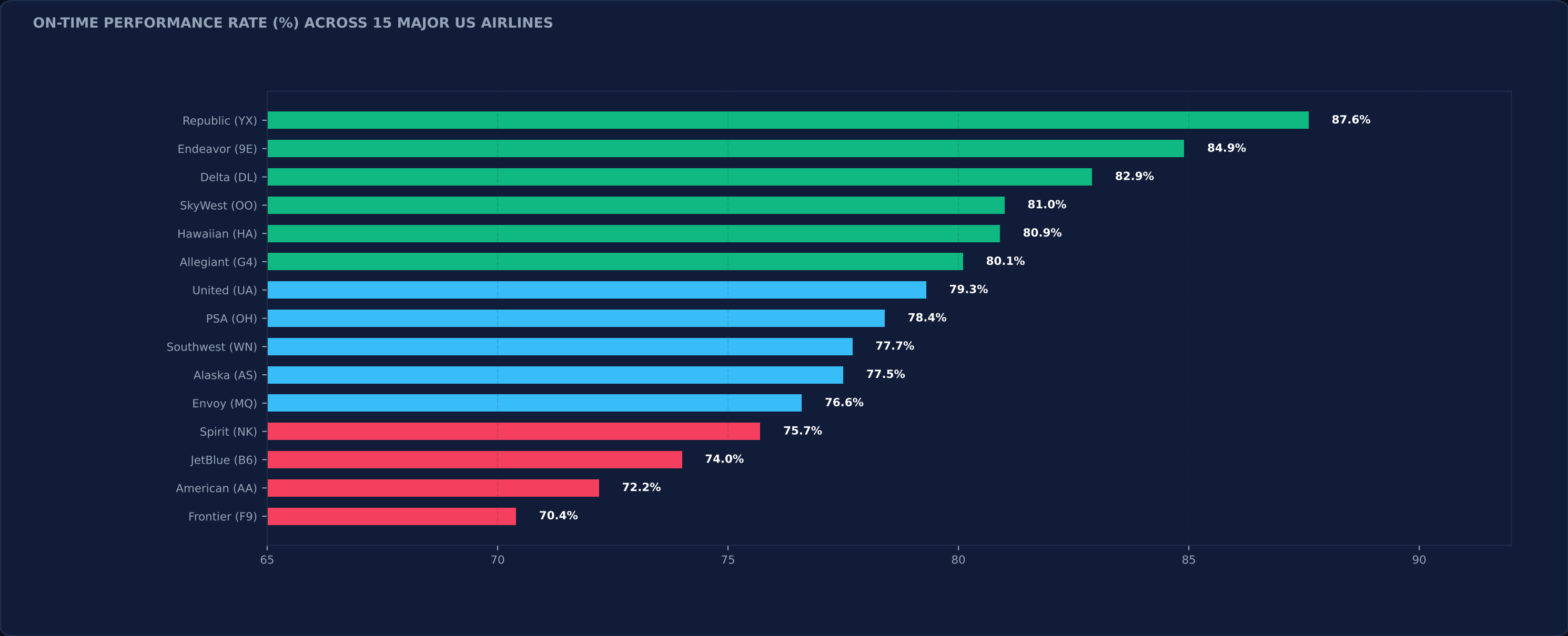
American (72.2%)

60,168 Flights Monitored

LOWEST ULCC PERFORMER

Frontier (70.4%)

High Utilization Vulnerability



TOTAL CANCELLATIONS

103,420

1.48% System Cancellation Rate

CARRIER / MECHANICAL ABORTS

44.2%

Code A: Technical Groundings

SEVERE WEATHER GROUNDINGS

38.8%

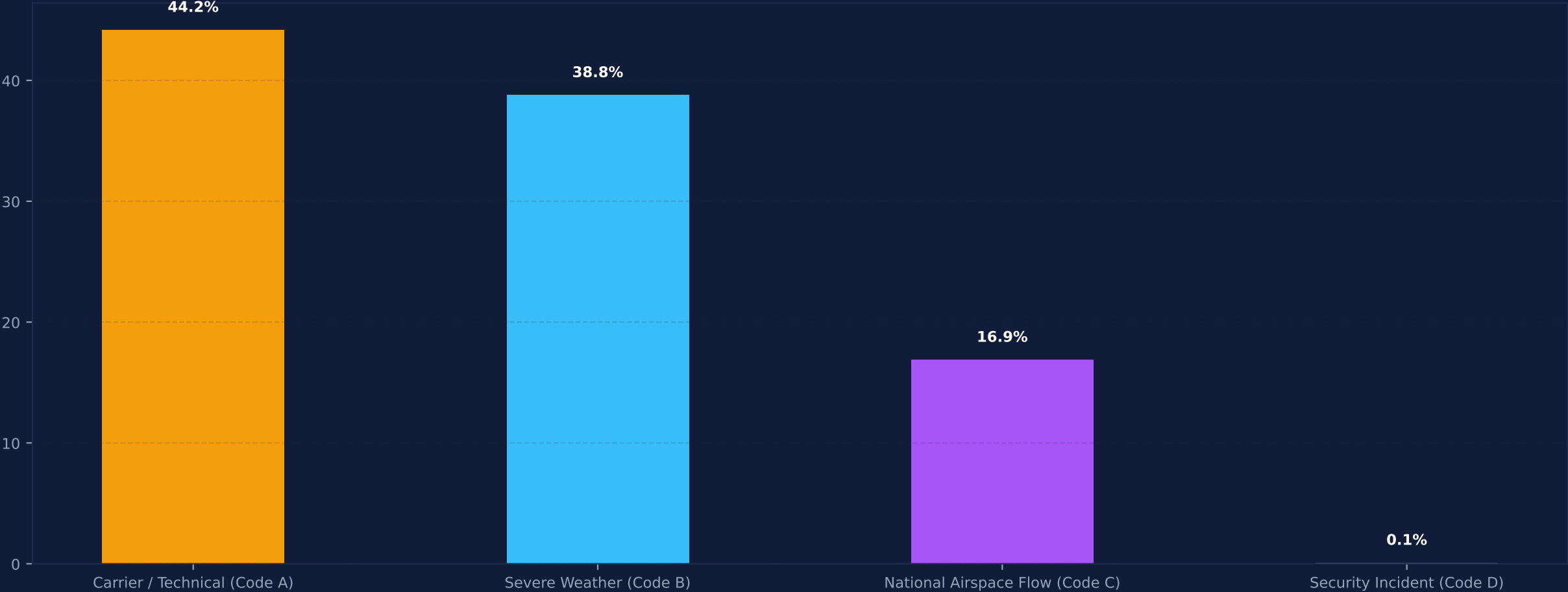
Code B: Winter & Convective

AIR TRAFFIC MANAGEMENT

16.9%

Code C: NAS Flow Control

FAA CANCELLATION CAUSE CODE SHARE (%)



FLEET AVG TAXI-OUT

17.4 min

Gate to Takeoff Roll

FLEET AVG TAXI-IN

7.6 min

Touchdown to Gate Arrival

SEVERE SURFACE HOLDS (>40M)

4.8%

340,000 Gridlocked Flights

EXCESS FUEL BURN

14.2M gal

~\$4.2M Ground Idling Penalty

TAXI-OUT DURATION FREQUENCY DISTRIBUTION (% OF ALL DEPARTURES)

